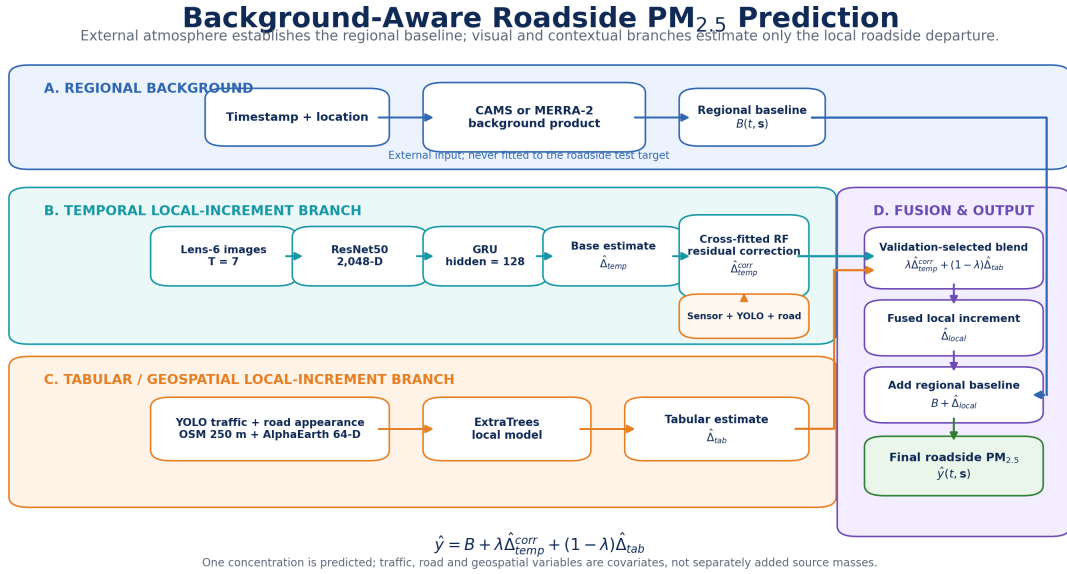


INTERNSHIP PROJECT REPORT

Background-Anchored Multimodal Modelling of Roadside PM_{2.5}

Roadside PM Visual Intelligence



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Internship period: May–July 2026

Report date: July 2026

Project repository: [Kumaryan12/roadside-pm-visual-intelligence](https://github.com/Kumaryan12/roadside-pm-visual-intelligence)

Declaration

I declare that this report is an accurate account of the engineering and research work performed in the *Roadside PM Visual Intelligence* codebase during the stated internship period. Results are reported with their validation protocol and known limitations. High random-window scores are explicitly separated from whole-date generalisation tests. External atmospheric products are identified as model or reanalysis estimates rather than roadside reference measurements. The source attribution experiments are described as exploratory particle or source proxies and are not presented as validated chemical apportionment.

Student signature: _____ **Date:** _____

Acknowledgements

I express my sincere gratitude to Prof. Raghunathan Rengaswamy for his guidance, technical questions and insistence on a physically defensible interpretation of the modelling problem. I am especially grateful to Dr. Sreejith Chakrapani, Postdoctoral Researcher at the Indian Institute of Technology Madras, and Navaneethakrishnan V, Project Associate and M.S. by Research scholar at the Indian Institute of Technology Madras, for their generous technical assistance, practical guidance and continued support throughout this project. I thank the data collection team and the contributors to the TRAQID and MUMMA datasets for making this work possible. I also acknowledge the open-source communities behind PyTorch, scikit-learn, OpenStreetMap, Ultralytics, Hugging Face, Copernicus CAMS, NASA MERRA-2 and the geospatial data services used in this project. I am grateful to Indian Institute of Technology Madras for hosting the internship and to National Institute of Technology Goa for supporting the academic programme.

Abstract

This internship developed and audited a reproducible multimodal system for predicting roadside $\text{PM}_{2.5}$ from a moving platform. The project began with a high-scoring image-sequence model on the historical TRAQID dataset and evolved into a more scientifically defensible five-day MUMMA study. The final data pipeline aligned 6,873 ten-second sensor observations from 55 runs with 20,619 preprocessed images from three camera lenses. It extracted 58 YOLO traffic features, 45 road-appearance features, 25 OpenStreetMap context features, 64 AlphaEarth embedding dimensions and 24 non-PM sensor, meteorology, time and mobility variables. External CAMS and MERRA-2 atmospheric products were added to separate regional background pollution from the local roadside increment.

The work compared direct tabular regression, frozen-CNN embeddings, GRU and LSTM sequence models, residual correction, background-plus-local-increment models, nested temporal-tabular ensembles, feature selection and early image-atmosphere fusion. A central finding was that random splitting of overlapping $T=7$ sequences produced optimistic scores: 98.32% of five-day test targets had already appeared as training context. Under whole-date holdout, direct image models generalised poorly. CAMS background decomposition improved the pooled five-day result to R^2 approximately 0.46. Replacing CAMS with MERRA-2 further improved the validation-selected nested ensemble to MAE $22.58 \text{ } \mu\text{g m}^{-3}$, RMSE $35.69 \text{ } \mu\text{g m}^{-3}$ and pooled R^2 0.538. A fixed 50:50 ensemble reached pooled R^2 0.566 but was retained only as a diagnostic because its weight was not validation-selected. Mean per-date R^2 remained negative, showing that new-day stability is not yet established.

The internship also explored particle-size regimes, NMF source-proxy components, assumption-conditioned source scenarios, spatial variograms, feature relevance, extreme-event audits and conformal uncertainty. These analyses established clear limits: chemical source apportionment requires differential OPC bins and chemical reference evidence, while future deployment claims require a frozen model evaluated once on untouched dates. The principal outcome is therefore both an improved model and a rigorous validation framework that distinguishes diagnostic accuracy from real generalisation.

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Introduction

1.1 Motivation

Roadside $\text{PM}_{2.5}$ varies over multiple spatial and temporal scales. A moving sensor observes the sum of a broad atmospheric background, neighbourhood-scale land-use effects and short-lived local increments from traffic, resuspension, road activity and meteorological mixing. A camera can observe traffic and road context, but it cannot directly measure aerosol mass. Conversely, a regional atmospheric product can describe the broad pollution regime but cannot resolve the exact plume or street-level exposure beside a moving vehicle. The project therefore treated roadside prediction as a multimodal inference problem rather than a pure image-regression task.

1.2 Original source-aware research direction

The internship did not begin with the final architecture already fixed. The original question was whether roadside particle behaviour—including a provisional effective-density proxy—could be explained using traffic, road condition, meteorology, gases and geospatial context:

$$\rho_{\text{eff}} = f(\text{traffic, road, weather, gases, OSM}). \quad (1.1)$$

Candidate mechanisms included exhaust, brake and tyre wear, road-dust resuspension, secondary aerosol and mixtures of these processes. The available measurements did not contain chemically speciated source contributions. Consequently, the project adopted the terms *source-aware feature*, *source proxy*, *particle-regime indicator* and *local pollution indicator*; it does not present exact chemical source apportionment.

The technical and scientific sequence was:

source-aware modelling $\rightarrow$ visual feature extraction $\rightarrow$ $\text{PM}_{2.5}$ prediction
 $\rightarrow$ validation audit $\rightarrow$ atmospheric background decomposition.

This evolution is central to the report. Later validation evidence and supervisor feedback changed the interpretation of earlier models instead of being retrofitted to a predetermined narrative.

The work also confronted a validation problem. Consecutive frames sampled every ten seconds are strongly autocorrelated. A $T=7$ sliding window shares six frames with the next window. If windows are constructed first and then randomly shuffled, nearly identical context enters both training and test partitions. The resulting score may be useful as a software or same-distribution diagnostic, but it is not evidence that the model will generalise to a new date. Resolving this distinction became a major scientific contribution of the internship.

1.3 Project objectives

The project had five linked objectives:

1. build a reproducible pipeline from raw sensor files and multi-lens videos to aligned modelling records;
2. derive interpretable traffic, road, sensor, geospatial and atmospheric predictors without using PM or OPC target proxies as reportable inputs;
3. compare image-only, tabular, temporal, residual and background-decomposition architectures;
4. establish leakage audits and whole-date evaluation as the principal generalisation test; and
5. explore particle-size and source-proxy analyses while preserving the distinction between predictive association and chemical source apportionment.

1.4 Main contributions

The internship produced the following concrete outputs:

- a five-day canonical MUMMA table with 6,873 aligned observations;
- 20,619 preprocessed images and a manifest enforcing exactly one image per retained lens and sample;
- complete YOLO traffic, road segmentation, OSM, AlphaEarth and atmospheric feature pipelines;
- an IDD-fine-tuned YOLO11m traffic detector and an Indian-road-tuned SegFormer model with IoU 0.893 and Dice 0.939;
- controlled road-area and white-balance experiments, including a 4 m^2 patch recovered as approximately 3.72 m^2 ;
- reusable frozen-CNN, GRU/LSTM, classical regression, residual correction and nested ensemble implementations;
- explicit overlap audits for random temporal splits;
- whole-date and nested validation protocols;
- a MERRA-2 background-plus-local-increment candidate with pooled whole-date R^2 0.538 under validation-selected fusion;
- uncertainty, feature relevance, variogram and sudden-event diagnostics; and
- consolidated experiment and architecture registries for reproducibility.

End-to-end roadside PM intelligence workflow



Figure 1.1: End-to-end workflow developed during the internship. Each stage writes a manifest, feature table, model artifact or audit file so that downstream results can be traced to their source.

1.5 Report scope and terminology

This report covers the initial 281-row synchronization and modelling pilot, the historical TRAQID reproduction, the two-date MUMMA development set, the complete five-day MUMMA experiments, the newer CAMS and MERRA-2 background studies, architecture-2 early fusion, uncertainty and exploratory particle/source work. In this report:

- *random diagnostic* means a split in which neighbouring rows or overlapping sequences can occur in different partitions;
- *fair* means that complete dates are withheld from model fitting;
- *pooled R^2* is computed after concatenating predictions from all held-out dates;
- *mean fold R^2* is the arithmetic mean of the five date-specific R^2 values; and
- *source proxy* denotes a latent or assumption-conditioned component, not a chemically validated source contribution.

Scientific and Technical Background

2.1 A scale-decomposition view of roadside PM_{2.5}

The final architecture is motivated by a simple mass-concentration decomposition:

$$y(t, \mathbf{s}) = B(t, \mathbf{s}) + \Delta_{\text{local}}(t, \mathbf{s}) + \epsilon, \quad (2.1)$$

where y is the roadside sensor concentration, B is the broad atmospheric background, Δ_{local} is the local departure from that background and ϵ includes instrument noise, alignment error and unmodelled processes. The terms are not separate physical samples collected by this project. They are a modelling decomposition. The external product supplies B ; the machine-learning branches estimate Δ_{local} from non-PM predictors.

This decomposition explains why traffic, road and land-use features remain useful even when images and atmospheric products are present. The atmospheric product is too coarse to resolve near-road variability. A generic image embedding contains visual appearance but does not explicitly guarantee stable vehicle counts, road texture or land-use context. Engineered variables expose these factors to a low-data tabular model and make the correction more auditable. They are predictive covariates, not separate mass contributions that are arithmetically added.

2.2 CAMS, MERRA-2 and local observations

The Copernicus Atmosphere Monitoring Service (CAMS) combines atmospheric modelling, emissions, meteorology and data assimilation to provide gridded aerosol and gas fields. MERRA-2 is a NASA atmospheric reanalysis that assimilates observations into a global numerical model and provides aerosol species and meteorological context. Neither product is a roadside reference monitor, and neither simply assigns a satellite pixel’s PM value to an image. They provide a regional estimate for a grid cell and time. The project aligned these external estimates to each roadside sample by timestamp and representative location.

In the earlier CAMS implementation, a small number of distinct time-grid values were repeated across many ten-second rows because the drive occupied a small region relative to the atmospheric grid and CAMS changed at a much coarser time step. MERRA-2 supplied more distinct matched atmospheric records and, in this dataset, proved a stronger baseline for the local-increment model. The background is available at inference time and must therefore also be supplied for an unseen date; it is not merely a training or tuning variable.

2.3 Images, CNN embeddings and temporal models

A convolutional neural network maps each frame to a feature vector encoding visual patterns. Frozen ImageNet backbones were used to reduce overfitting in the small five-day dataset. ResNet50 produced a 2,048-dimensional global-average-pooled embedding. ConvNeXt-tiny, MobileNetV2, EfficientNet-B0 and VGG16 representations were also supported or tested.

A recurrent neural network receives a sequence of embeddings $\mathbf{x}_{t-T+1}, \dots, \mathbf{x}_t$ and produces a temporal representation:

$$\mathbf{h}_t = \text{GRU}(\mathbf{x}_{t-T+1:t}), \quad \hat{y}_t = g(\mathbf{h}_t). \quad (2.2)$$

GRUs and LSTMs can capture persistence and evolving visual context. However, the same persistence makes random splitting dangerous: overlapping windows can share raw frames across partitions.

2.4 Tree ensembles and residual learning

Random Forest averages decorrelated decision trees trained on bootstrap samples. ExtraTrees further randomises split thresholds, often producing a strong nonlinear model for mixed tabular variables. HistGradientBoosting adds shallow trees sequentially to reduce residual error. Ridge provides a regularised linear reference.

Residual learning begins with a base prediction $\hat{y}_{\text{base}}$ and fits a second model to

$$r = y - \hat{y}_{\text{base}}. \quad (2.3)$$

The corrected prediction is $\hat{y} = \hat{y}_{\text{base}} + \hat{r}$. To prevent the residual model from learning errors caused by in-sample base predictions, the project used out-of-fold or cross-fitted base predictions wherever possible.

2.5 Evaluation metrics

The report uses:

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|, \quad (2.4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}, \quad (2.5)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (2.6)$$

MAE measures typical absolute error; RMSE penalises extreme misses more strongly. R^2 compares squared error with a mean-prediction baseline. A negative date-specific R^2 means the model was worse on that date than predicting that date's mean. Pooled R^2 can still be positive if the model captures differences between daily means, even when within-day fluctuations remain difficult.

Datasets and Data Engineering

3.1 Initial synchronized sensor–image pilot

The first substantial engineering task was constructing a synchronized dataset, not training a neural network. The early MUMMA pilot contained 281 sensor rows, 252 rows with usable corresponding images and 756 successfully preprocessed frames from lenses 1, 4 and 6. Sensor timestamps and video run metadata were used to locate the corresponding video offset. Frames were then extracted, rotated into a common orientation, cropped, masked to remove the platform and assigned persistent sample and run identifiers. Raw videos were never modified.

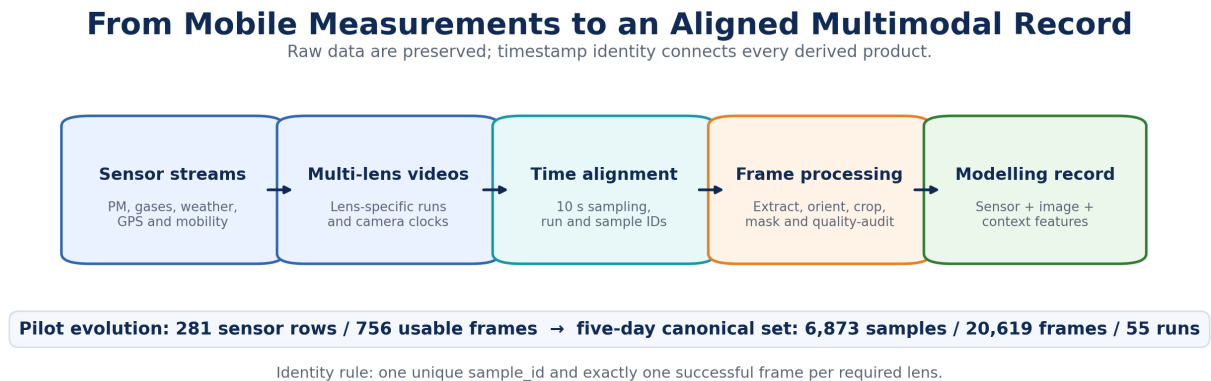


Figure 3.1: Evolution of the synchronized data pipeline from the compact pilot to the canonical five-day dataset.

This pilot established the invariant used throughout the later project: sensor, image, traffic, road and geospatial data must join through explicit identity and timestamp fields rather than through implicit row ordering. Later orientation audits replaced lens 4 with lens 2 in the canonical three-view configuration, yielding lenses 1, 2 and 6.

3.2 TRAQID historical benchmark

The historical benchmark contained 26,558 $T=7$ front/rear image sequences. The presentation reproduction randomly assigned 21,246 sequences to training/validation and 5,312 to test. Because overlapping windows were shuffled, the experiment reproduced a high same-distribution score but was not a future-date evaluation. TRAQID remained valuable for verifying architecture code, residual learning and later CAMS/MERRA comparisons.

3.3 MUMMA two-date development dataset

The subsequent two-date development set used 2,167 aligned records from 1 and 3 February 2026. It established:

- SSD ingestion without modifying the raw external data;
- ten-second sensor aggregation;
- multi-lens video discovery and timestamp alignment;
- frame extraction and lens-specific orientation/cropping;
- YOLO, road, CNN embedding and sequence generation;
- direct and residual modelling; and
- the first explicit overlap audit.

Lenses 1, 2 and 6 were retained after inspecting orientation and usable road context.

3.4 Canonical five-day MUMMA dataset

The complete dataset covers 1–5 February 2026. It contains 6,873 aligned ten-second samples from 55 runs and exactly three preprocessed frames per sample. Table 3.1 summarises the data.

Table 3.1: Five-day MUMMA coverage. Time bounds include gaps between separate runs and should not be interpreted as continuous driving duration.

Date	Rows	Runs	First timestamp	Last timestamp
2026-02-01	1,265	10	08:27:06	14:57:47
2026-02-02	1,736	15	09:12:52	17:15:34
2026-02-03	902	4	10:58:02	14:51:19
2026-02-04	1,542	16	08:58:49	14:06:01
2026-02-05	1,428	10	08:33:40	13:31:57
Total	6,873	55	Five collection dates	

Canonical five-day MUMMA dataset: 6,873 samples, 55 runs, 20,619 images

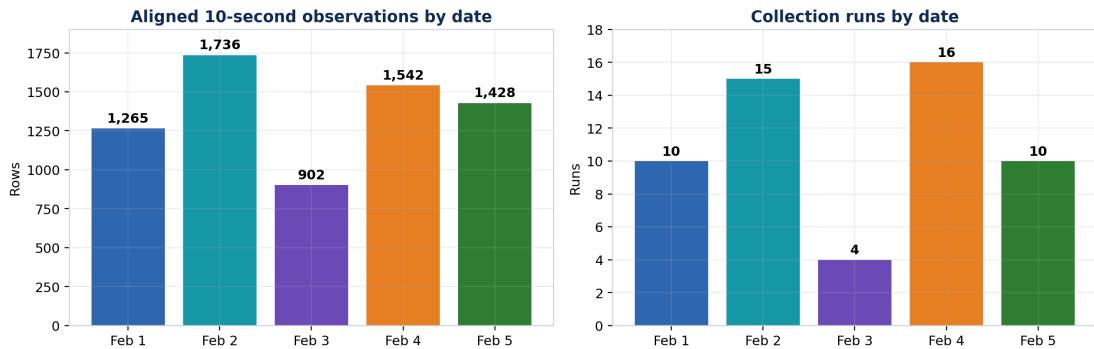


Figure 3.2: Distribution of aligned observations and collection runs across the five dates.

3.5 Alignment and identity policy

The canonical merge enforced a strict identity policy:

1. `sample_id` is unique at the sensor-record level;
2. each sample has exactly one successful frame for lenses 1, 2 and 6;
3. sequences are grouped by `run_id`;
4. fair sequences cannot cross a run or date boundary; and
5. all feature tables join through explicit sample identifiers rather than row position.

The combined manifest contained 20,619 frame rows, all successfully preprocessed. The raw SSD remained unchanged; processed outputs were written to the project artifact tree.



Figure 3.3: Example of the three retained preprocessed camera views for one aligned sample. Different views provide complementary road, sky and traffic context; they are not assumed to be interchangeable.

3.6 Target and predictor safety

The primary target was the sensor field `sPM2`. The reportable predictor groups exclude `sPM1`, `sPM2`, `sPM4`, `sPM10`, OPC number channels and `sTPS`. These channels were retained only for target construction, quality checks and exploratory particle-regime analysis. This separation prevents direct PM or near-equivalent OPC leakage into the roadside PM_{2.5} predictor.

Feature Engineering

4.1 Feature-family overview

The authoritative feature membership is stored in `artifacts/runs/mumma_5day_complete_nonpm_models_v1/feature_groups.json`. Table 4.1 summarises the principal families. Features were generated once and then reused across model comparisons to avoid silent changes between experiments.

Table 4.1: Canonical non-PM feature families.

Family	Count	Content
Sensor/met/gas/time/mobility	24	Temperature, relative humidity, CO ₂ , VOC/NO _x indices, gases, cyclic time, GPS, speed, course and HDOP.
YOLO traffic	58	Vehicle class counts, heavy/motor totals, confidence and box geometry, occupancy, exhaust/resuspension proxies and per-lens variants.
Road appearance	45	SegFormer road area, colour, illumination, edge, texture and haze descriptors, aggregated and per road-facing lens.
Road depth/metric area	16	Depth percentiles, mask-cleaning diagnostics and estimated road area; provisional.
OSM	25	Activity/POI counts and road-network hierarchy within 250 m.
AlphaEarth	64	Annual latent embedding dimensions A00–A63.

4.2 Sensor, meteorology, gases, time and mobility

The 24 variables were:

```
temp, rh, k30Co2, sTemp, sRh, sVocI, sNoxI, co_ppb, no2_ppb, so2_ppb, o3_ppb_compensated, ch4_ratio,
nh3_ppm, hour_sin, hour_cos, minute_sin, minute_cos, lat, long, alt, sog_clean, cog_sin, cog_cos, hdop.
```

Cyclic encoding prevents an artificial discontinuity between, for example, 23:59 and 00:00. Course was encoded with sine and cosine for the same reason. Latitude and longitude were retained as predictive location variables but were recognised as a potential route-memory mechanism in random splits.

4.3 YOLO traffic features

YOLO11m was fine-tuned on the IDD 15-class Indian-road dataset and then applied to all 20,619 preprocessed frames. The extraction completed successfully for every frame and produced 136,534 PM-relevant road-user detections. The object classes were mapped to motorcycle, car, auto-rickshaw, truck, bus, bicycle and an unknown-vehicle fallback.

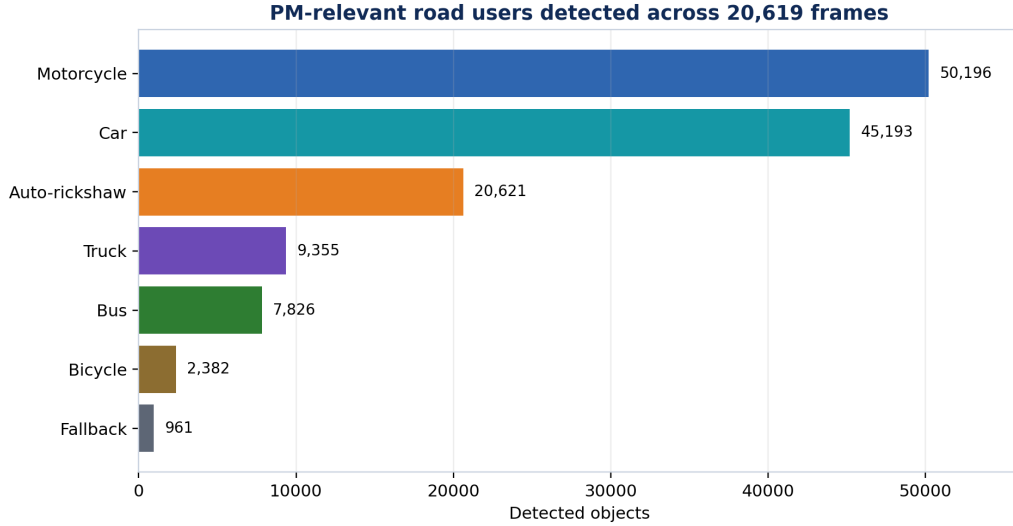


Figure 4.1: Detected PM-relevant road users in the five-day MUMMA dataset. Counts are observations in images, not unique physical vehicles.

The 58 model features comprise:

- aggregate counts for total vehicles, heavy vehicles, motor vehicles and each mapped class;
- initial exhaust and resuspension proxy scores based on the detected mix;
- mean and maximum bounding-box area, average confidence and maximum confidence;
- the same core counts, proxies and geometry for lenses 1, 2 and 6; and
- `yolo_lens_count` as a view-availability audit.

These variables do not claim to measure emitted mass. They describe source opportunity, traffic composition, proximity/occupancy and activity. A vehicle count can correlate negatively in a global table because routes, background pollution and time of day confound the association. Within-day centring changed some vehicle correlations from negative to positive, demonstrating why correlation is not a causal emission factor.

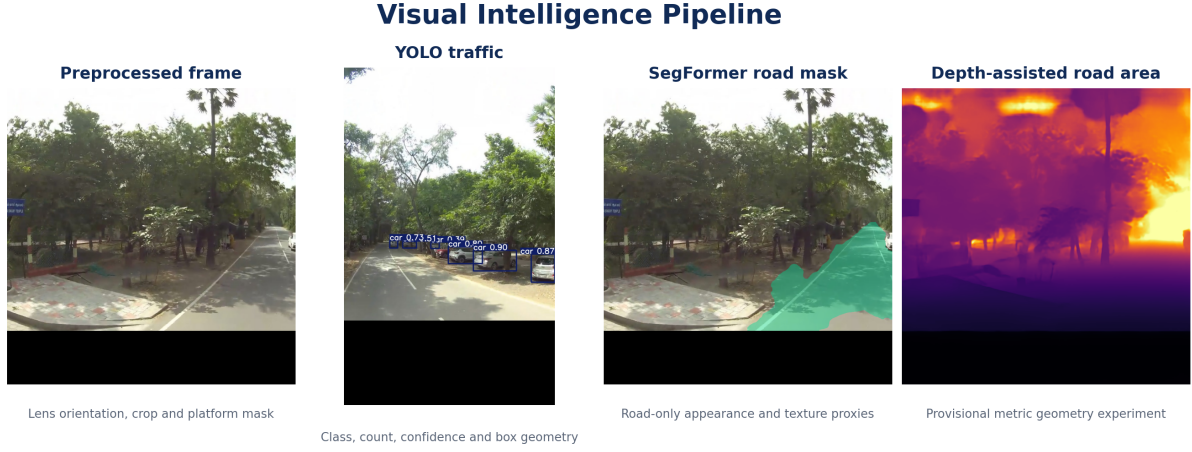


Figure 4.2: Representative visual processing outputs: calibrated frame, IDD traffic detections, SegFormer road overlay and depth-assisted geometry.

4.4 Road appearance features

Road extraction initially used OpenCV colour/texture thresholds and CLIP-based semantic experiments. These approaches were unstable under shadows, glare, road-colour variation, occlusion and complex Indian scenes. The project therefore moved to a Cityscapes-pretrained SegFormer model, which was fine-tuned for binary Indian-road segmentation. Its independently evaluated road-mask performance was approximately IoU 0.893 and Dice 0.939. This was one of the strongest component-level computer-vision results of the internship.

The SegFormer-B0 road model was run on the calibrated road-facing lenses 1 and 6. Of 13,746 input frames, 13,041 succeeded and 705 failed. The mean segmented road-area ratio among successes was 0.1163.

Eleven underlying measurements were used:

1. road area ratio;
2. mean brightness;
3. mean saturation;
4. contrast standard deviation;
5. shadow-pixel ratio;
6. glare-pixel ratio;
7. brown-pixel ratio;
8. gray/dry-pixel ratio;
9. edge density;
10. Laplacian texture standard deviation; and
11. haze-flatness proxy.

Each measurement was represented as an all-lens mean, all-lens maximum, lens-1 value and lens-6 value, with an additional road-lens count. These variables describe visible road state and surface opportunity for resuspension; they do not measure the amount of road dust suspended in air.

4.5 Depth-derived road area

A depth-assisted procedure attempted to convert the road mask to an estimated metric area. For road pixel (u, v) with estimated metric depth z , camera intrinsics mapped the pixel into 3D:

$$x = \frac{(u - c_x)z}{f_x}, \quad y = \frac{(v - c_y)z}{f_y}. \quad (4.1)$$

Neighbouring road points were connected as triangles and their areas summed. Occluded road beneath vehicles was then explored through three successive heuristics. Version 1 used a vehicle-footprint prior, box visibility, road-contact confidence and conservative scaling. Removing road-contact confidence in version 2 overestimated distant vehicles. Version 3 therefore introduced depth gating:

$$A_{\text{hidden}} = A_{\text{footprint}} V_{\text{bbox}} D_{\text{factor}} C_{\text{conservative}}. \quad (4.2)$$

A controlled $2\text{ m} \times 2\text{ m}$ marked patch had true area 4.00 m^2 ; the pipeline estimated approximately 3.72 m^2 , an error of about 6.94%. Despite that promising controlled check, the complete five-day extraction remained provisional. It produced a value for all 6,873 samples, with mean 89.48 m^2 , but every result was flagged either `needs_visual_review` or `bad_low_road_mask_area`. The cleaning stage removed a median 54.8% of the initial mask. Models using this family generally degraded. It is retained as a documented negative result, not as a trusted physical road-area measurement.

4.6 White-balance calibration

Road colour and dryness proxies are sensitive to illumination and camera colour processing. Automatic, daylight and fixed-6000 K white balance were compared through the coefficient of variation of the road-brown feature. The values were 27.53%, 25.44% and 12.17%, respectively. Fixed 6000 K was selected because it was the most repeatable tested setting, not because it establishes an objectively correct physical road colour.

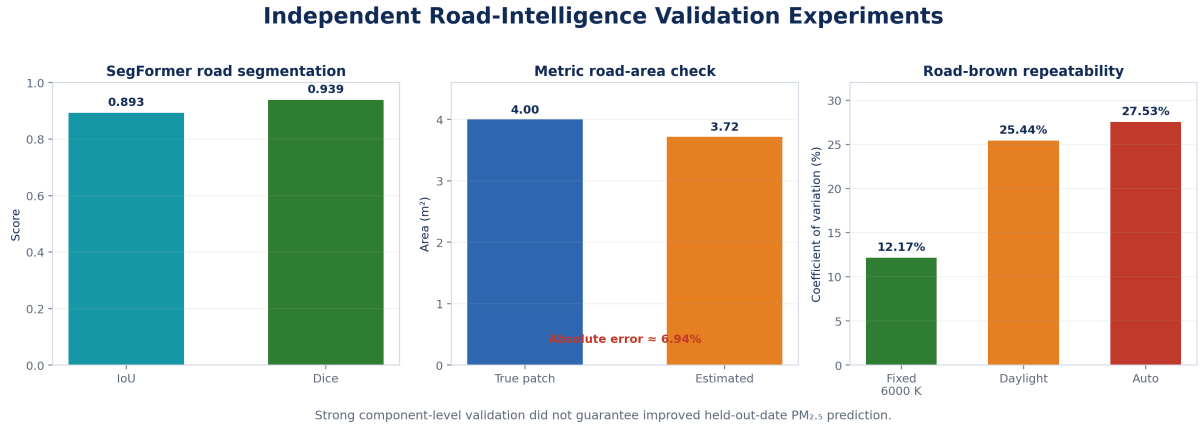


Figure 4.3: Controlled validation of road segmentation, metric area and white-balance repeatability. Better component quality did not automatically produce better held-out-date PM_{2.5} prediction.

4.7 OpenStreetMap context

OpenStreetMap queries were cached by corridor tile and summarised locally inside a 250 m radius. The final run reused 41 cached tiles, fetched one remaining tile and wrote complete

features for all 6,873 samples. The 25 predictors were:

- **activity/POI counts (14):** fuel station, restaurant, bus stop, parking, construction, industrial, factory, warehouse, marketplace, park, school/college, hospital, commercial and retail;
- **road context (11):** road-segment count, total road length, and counts of motorway, trunk, primary, secondary, tertiary, residential, service, living-street and unclassified roads.

Rounded coordinates and OSM location keys were join metadata and were not counted as predictors. OSM provides slowly varying context about potential activity and road hierarchy. It cannot describe the instantaneous traffic state.

4.8 AlphaEarth embeddings

AlphaEarth supplied a 64-dimensional annual embedding for every location. The dimensions A00–A63 encode learned geospatial context, but their individual physical meanings are latent. They must not be renamed as specific land-cover types without an independent interpretation study. AlphaEarth A36 was the only dimension selected in every nested feature-selection fold, while several other dimensions showed repeated relevance.

4.9 Atmospheric background and meteorological fields

Atmospheric experiments used matched CAMS, MERRA-2 and ERA5 fields, including background PM estimates, aerosol components, temperature, dew point, humidity, wind, pressure, precipitation and solar radiation where available. The current background-plus-local architecture uses the external PM estimate as B in Equation 2.1. Architecture 2 instead treats 22 atmospheric fields as ordinary early-fusion predictors.

4.10 Composite feature groups

Table 4.2: Composite feature groups used in the principal MUMMA experiments.

Group	Predictors
visual_yolo_road	103
sensor_plus_visual	127
visual_yolo_road_area_depth	119
sensor_plus_visual_area_depth	143
visual_yolo_road_osm	128
sensor_plus_visual_osm	152
sensor_plus_alphaearth	88
visual_yolo_road_alphaearth	167
sensor_plus_visual_alphaearth	191
visual_yolo_road_osm_alphaearth	192
sensor_plus_visual_osm_alphaearth	216

Scientific Reframing After Supervisor Review

5.1 Why generic residual correction was conceptually weak

The earlier architecture predicted total roadside $PM_{2.5}$ with a CNN-RNN and then trained a tree model on

$$r_{\text{old}} = PM_{\text{observed}} - \widehat{PM}_{\text{image base}}. \quad (5.1)$$

This is mathematically valid, but the residual has no single physical meaning. It can contain regional background, image-model error, traffic, resuspension, meteorology, sensor noise, date shift and synchronization error. The model therefore learned to “correct whatever the image model missed” rather than answering which local roadside conditions explained the departure.

5.2 Supervisor’s physical question and identifiability

Prof. Raghunathan Rengaswamy asked whether traffic, road and image variables were closing a scientifically meaningful particulate balance or merely acting as generic predictors. A useful conceptual relation is

$$PM_{\text{atmospheric}} \approx PM_{\text{background}} + PM_{\text{road}} + PM_{\text{vehicle}} + PM_{\text{other}} + \epsilon. \quad (5.2)$$

The project did not observe reference values for the road and vehicle terms. The same total increment can be explained by many combinations; for example, $30 = 20 + 10 = 5 + 25$. Exact source contributions are therefore not identifiable without chemical or source-specific measurements.

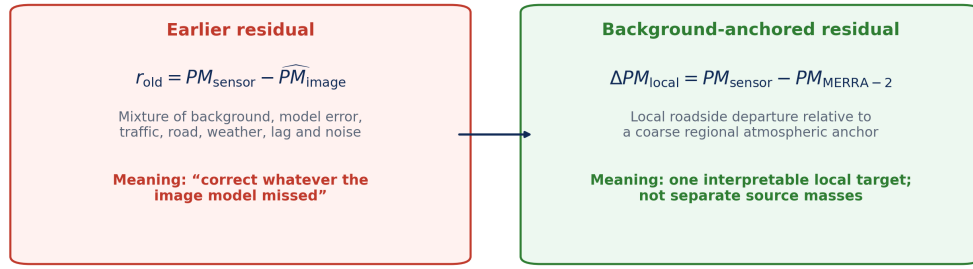
5.3 Physically structured target

The revised target is

$$\Delta PM_{\text{local},t} = PM_{\text{sensor},t} - PM_{\text{background},t}. \quad (5.3)$$

For the strongest current experiment, MERRA-2 supplies the external background. Traffic, road, meteorology, images and geospatial context jointly predict one local roadside increment. Negative increments are retained because they may represent cleaner local conditions, background mismatch, spatial-resolution differences or measurement uncertainty. The safest interpretation is *local roadside increment relative to MERRA-2 background*, not traffic pollution alone.

Scientific Reframing: From Generic Error Correction to a Physical Scale Decomposition



Identifiability remains: traffic and road variables explain the local increment jointly; they cannot yield exact source percentages without chemical reference data.

Figure 5.1: Conceptual transition from an unrestricted image-model error to a background-anchored local target.

5.4 Relation to the proposed image–satellite architecture

The later handwritten proposal suggested learning atmospheric $PM_{2.5}$ from images and satellite/reanalysis inputs, followed by interpretation using road, vehicle, OSM and uncertainty information. This was tested as Architecture 2: target-frame image embeddings and 22 atmospheric variables were fused in a single ExtraTrees model. Because the training target was still the roadside sensor, that experiment estimated roadside $PM_{2.5}$, not pure atmospheric $PM_{2.5}$. Its results are reported later. The final recommended formulation uses MERRA-2 as an external anchor and reserves detailed traffic and road variables for the local-increment branch.

Model Architectures

6.1 Architecture inventory

The codebase contains a broad experimental ladder rather than one monolithic model. Table 6.1 groups the implemented designs.

Table 6.1: Architecture families implemented or audited during the project.

ID	Family	Purpose
FE-01/02	Frozen or supervised CNN	ResNet50, ConvNeXt, MobileNetV2, EfficientNet-B0 and VGG16 image encoders/regressors.
FE-03	YOLO11m	IDD traffic detection and interpretable traffic features.
FE-04	SegFormer-B0	Road segmentation and appearance features.
FE-05	Depth road area	Provisional metric road-area estimation.
FE-06/07	AlphaEarth and OSM	Learned geospatial embedding and interpretable 250 m context.
PM-01	Classical tabular suite	Ridge, Random Forest, ExtraTrees and Hist-GradientBoosting direct prediction.
PM-02/03	CNN + classical/MLP	Single-frame embedding models and PCA regressors.
PM-04	Generic GRU/LSTM	Frozen CNN embedding sequences with a recurrent prediction head.
PM-05/06	Historical CNN-LSTM	HVAQ and TRAQID paper-style sequence reproduction.
PM-07–11	Multimodal fusion RNNs	Early concatenation, late fusion, gated dual-branch and gated multi-view temporal models.
PM-12	OOF GRU + clipped residual	Historical TRAQID presentation reproduction.
PM-13	Image base + tabular residual	Strict base/correction architecture used in early MUMMA experiments.
PM-14	Background + local increment	External CAMS/MERRA background with a learned roadside departure.
PM-15	Nested temporal-tabular ensemble	Validation-selected fusion of temporal and tabular local-increment branches.

ID	Family	Purpose
PM-16	Weighted/stacked ensembles	Diagnostic weighted, greedy and stacking combinations.
PM-17/18	Numerical forecasting	Multi-horizon LSTM and classical numerical baselines.
SA-01-03	Particle/source proxy	NMF, assumed-profile NNLS and derived particle-regime prediction.

6.2 Generic frozen-CNN GRU/LSTM

The principal sequence trainer used a one-layer unidirectional GRU or LSTM with hidden dimension 128. The last recurrent output passed through layer normalisation, dropout 0.3, a 128-unit ReLU layer, another dropout and a linear output. Training used AdamW with learning rate 10^{-4} , weight decay 10^{-4} , batch size 32, SmoothL1 loss with beta 1.0, up to 100 epochs and patience 15. Optional log-target transformation and date-balanced sampling were tested in robust variants.

Backbones included ResNet50, ConvNeXt-tiny and MobileNetV2. Views included lens 1, lens 2, lens 6, the mean of lenses 1/2/6 and concatenated lenses 1/2/6. Sequence lengths $T=3, 7$ and 9 were prominent; the early ladder also supported $T=1, 13$ and 31.

6.3 Historical TRAQID OOF residual architecture

The historical presentation architecture used:

1. front and rear ResNet50 global-average-pooled embeddings (2,048 dimensions each);
2. concatenation to 4,096 dimensions and projection to 512;
3. a two-layer unidirectional GRU with hidden size 256 and dropout 0.25;
4. three shuffled out-of-fold GRU models;
5. an OOF residual target $y - \hat{y}_{\text{OOF}}$;
6. 112 row-level weather, time, YOLO and road variables aggregated across $T=7$ by last, mean, standard deviation, minimum and maximum;
7. 558 residual predictors after adding base-prediction transformations;
8. conservative ExtraTrees with 900 trees, `max_features=0.35` and `min_samples_leaf=4`; and
9. residual clipping to $\pm 12.5 \mu\text{g m}^{-3}$.

This architecture delivered the historical R^2 0.966 result, but its random overlapping-window split made that result non-generalisation evidence.

6.4 Strict image base plus residual correction

In the first MUMMA architecture, the CNN-RNN predicted $\text{PM}_{2.5}$ directly. A tabular model then predicted the outer-validation residual using selected non-PM features. Candidate correction models were Ridge, Random Forest, ExtraTrees and HistGradientBoosting. The design

tested whether explicit traffic, road and sensor variables could correct information that the image sequence missed.

6.5 Background plus local increment

The scientific model evolved to:

$$\hat{y} = B + \hat{\Delta}_{\text{local}}, \quad (6.1)$$

where B is external CAMS or MERRA-2 background. The target for the learned branch is $y - B$, not total $\text{PM}_{2.5}$. This reduces the burden on the local model: it need not infer an unobserved regional baseline from a short roadside image sequence.

6.6 Nested temporal-tabular ensemble

The latest structured architecture has two local-increment branches:

- **temporal branch:** lens-6 ResNet50 embeddings, T=7 GRU and a cross-fitted Random Forest residual correction using sensor, YOLO and road variables;
- **tabular branch:** ExtraTrees trained on YOLO, road, OSM and AlphaEarth variables.

The branches are fused using a convex weight λ selected on validation RMSE:

$$\hat{y} = B + \lambda \hat{\Delta}_{\text{temp}}^{\text{corr}} + (1 - \lambda) \hat{\Delta}_{\text{tab}}. \quad (6.2)$$

The selected weight is frozen before outer-test prediction. A 50:50 average was also computed as a diagnostic; because it was not selected by the stated validation rule, it is not the primary model even when its pooled score is higher.

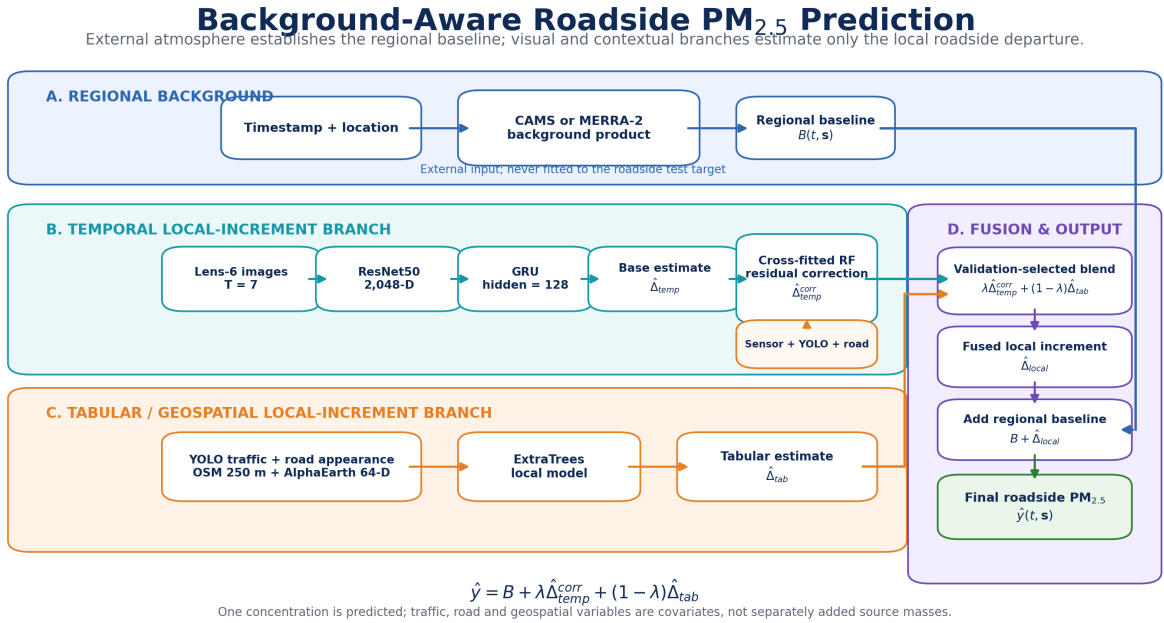


Figure 6.1: Current background-plus-local-increment architecture. Traffic, road, OSM and AlphaEarth are covariates for the local departure; their predictions are not separately added as physical source masses.

6.7 Architecture 2: image-atmosphere early fusion

The professor-proposed alternative combined a target-frame front/rear ResNet50 embedding with 22 atmospheric features. The image embedding was reduced by train-only randomised PCA (requested 64 components), concatenated with CAMS, MERRA-2 and ERA5 variables and passed to an ExtraTrees regressor. A 90% validation-absolute-residual conformal interval quantified uncertainty.

This tests predictive multimodal fusion, but it does not isolate “pure atmospheric PM” because the target remains the roadside sensor. It also does not estimate road and vehicle source contributions. The experiment was run on both TRAQID and MUMMA under complete-date holdout.

Experimental Design and Validation

7.1 Random row split

Rows are shuffled between training and test. Adjacent ten-second observations and the same dates occur in both partitions. This protocol answers: “Can the model interpolate among highly similar observations from the same collection distribution?” It does not answer: “Will the model work on a new date?”

7.2 Random sequence split after window construction

T=7 or T=9 windows are first built with stride one and then shuffled. Adjacent windows share most raw frames. For the five-day T=7 experiment:

- 6,451 of 6,873 raw samples appeared in multiple partitions;
- 5,142 raw samples overlapped training and test;
- 1,287 of 1,309 test target frames appeared as training context; and
- the target/context leakage fraction was 98.32%.

Why validation protocol changed the scientific conclusion

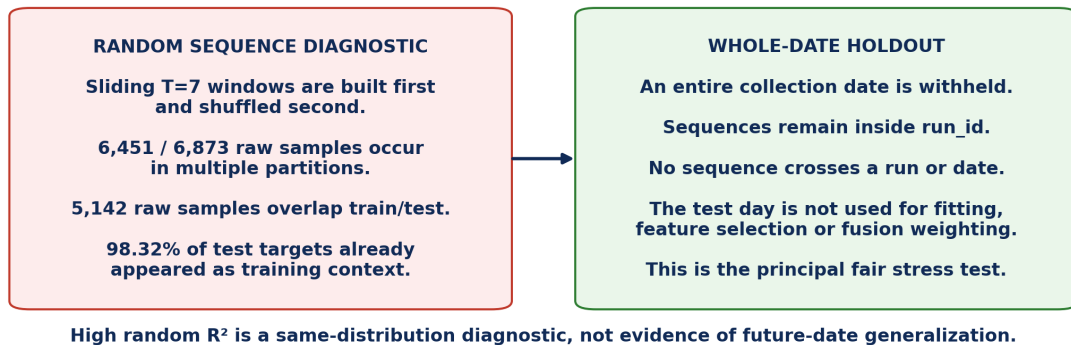


Figure 7.1: Contrast between the random overlapping-window diagnostic and the whole-date holdout used for fair evaluation.

7.3 Whole-date outer holdout

Each of the five dates is held out in turn. Training uses the other dates; sequences remain within `run_id`. Each row receives one outer-test prediction from a model that did not train on its date. Pooled metrics are computed over the concatenated predictions, while fold metrics reveal date-specific stability.

7.4 Nested selection and cross-fitting

For each outer fold, hyperparameters or fusion weights are chosen from an outer-validation partition only. Residual correction on the validation date is cross-fitted by run to avoid training and scoring the correction on the same residuals. The outer-test date is untouched until the architecture and weight are frozen.

Even this protocol remains *exploratory* at the project level because the same five dates informed multiple architecture decisions during two months of development. Confirmatory evidence requires new dates collected after the pipeline is frozen.

7.5 Pooled versus per-date R^2

Pooled R^2 uses one global mean:

$$R^2_{\text{pooled}} = 1 - \frac{\sum_d \sum_i (y_{di} - \hat{y}_{di})^2}{\sum_d \sum_i (y_{di} - \bar{y})^2}. \quad (7.1)$$

Its denominator includes between-date baseline variation. Mean fold R^2 instead computes R^2 within each day and averages the five values. A model can therefore have positive pooled R^2 by predicting high days versus low days while still performing poorly relative to each day's local mean. Both statistics are reported.

Scientific caution. No random-window result in this report is labelled as future-date generalisation. Whole-date results are fair stress tests, but final deployment claims still require untouched future collection dates.

PM_{2.5} Prediction Results

8.1 Historical TRAQID random-window progression

Table 8.1 reconstructs the result shown in the historical presentation. Each improvement was measured on the same overlap-contaminated random-window benchmark.

Table 8.1: Historical TRAQID random-window result progression.

Variant	MAE ($\mu\text{g m}^{-3}$)	RMSE ($\mu\text{g m}^{-3}$)	R^2
Single ResNet50-GRU	5.991	10.359	0.951
Tuned residual using base prediction	5.440	9.847	0.956
Three-model base ensemble	5.113	9.434	0.960
OOF ExtraTrees residual, unclipped	4.234	8.867	0.964
OOF ExtraTrees residual, clipped ± 12.5	4.212	8.717	0.966

Later TRAQID date-held-out ResNet50-GRU folds obtained R^2 0.249, -0.079, -0.883, -0.313 and -0.307. The gap between these values and 0.966 is evidence of date shift and overlap effects, not a contradiction in the implementation.

8.2 TRAQID single-image and engineered-feature baselines

Two standalone routes were also compared before atmospheric decomposition. A single front/rear ResNet50 image representation with an MLP achieved approximately R^2 0.49, whereas a Random Forest using roughly 112 engineered traffic, road, weather and time variables achieved approximately R^2 0.84 under the historical split. This established an important design lesson: for small roadside datasets, explicit traffic and road summaries can be more useful than generic ImageNet embeddings when used alone. It did not, however, resolve the later unseen-date validation problem.

8.3 TRAQID matched direct and background random tests

A later matched random experiment compared direct PM_{2.5}, CAMS plus local increment, MERRA-2 plus local increment and GEOS-CF. The same 3,983 test sequences and 714 correction features were used.

Table 8.2: Selected TRAQID matched random-sequence results. These remain random diagnostics.

Framework	Correction	MAE	RMSE	R^2
Direct PM _{2.5}	Random Forest	6.583	11.171	0.9413
CAMS + local increment	ExtraTrees	6.197	10.882	0.9443
MERRA-2 + local increment	Random Forest	5.948	10.657	0.9466
GEOS-CF + local increment	ExtraTrees	6.779	11.264	0.9403

MERRA-2 was best in the matched random comparison, although the uncorrected MERRA-2 image-base row had the lowest MAE (5.878) while its R^2 was slightly lower (0.9460). Metric rankings can differ because MAE and RMSE weight errors differently.

8.4 TRAQID whole-date background ensembles

The same background concept did not transfer as strongly to TRAQID whole-date holdout. The MERRA-2 fixed 50:50 temporal-tabular diagnostic was best with MAE 29.32, RMSE 43.18 and pooled R^2 0.147. Its mean fold R^2 was only 0.047. The validation-selected MERRA-2 ensemble reached pooled R^2 0.036. CAMS fixed 50:50 reached 0.076. These results demonstrate that an architecture can help one dataset and fail to provide the same benefit under a different route, sensor, background or date distribution.

8.5 MUMMA 281-row pilot modelling

The compact pilot modelling table contained 281 rows, about 190 original columns and 252 valid image alignments. Twenty-nine ResNet embeddings failed, and the observations represented only about 48 minutes of a volatile collection. Reportable predictors excluded all PM and OPC target proxies.

The best fair pilot used a meteorology/gas/time ExtraTrees base on a log-transformed target and an ExtraTrees residual model using selected road, traffic and OSM features. It obtained MAE 5.646 $\mu\text{g m}^{-3}$, RMSE 9.719 $\mu\text{g m}^{-3}$ and R^2 0.650. Small-data temporal approaches were weaker:

Table 8.3: Compact MUMMA pilot comparison.

Model	RMSE	R^2
Optimised non-temporal residual model	9.72	0.650
Hand-engineered T=7 model	12.09	0.466
Best ResNet temporal model	13.21	0.425
Best ResNet50-LSTM	14.27	0.329

The previous-row persistence baseline reached only about R^2 0.466 and most magnitude predictability disappeared by lag two. This pilot demonstrated that the dataset was too short and volatile for reliable deep temporal learning.

8.6 Two-day MUMMA prototyping

In random overlapping sequences, the image-base R^2 increased from 0.679 at T=3 to 0.797 at T=7 and 0.828 at T=9. Leakage increased at the same time. At T=9, 99.27% of test target frames had already appeared as training context. A causal trailing-three-row smoothed target produced MAE 3.875, RMSE 6.033 and R^2 0.958. This was a useful upper-bound experiment but not a deployment score because the target was smoother and the split was heavily contaminated.

The two-day whole-date test was much worse. The best T=7 visual correction had RMSE 41.00 and R^2 -0.542. With only two dates, there was no independent nested date for correction-model selection.

8.7 Five-day image backbone and view comparison

Table 8.4 reports mean performance over five held-out dates. Changing the CNN or recurrent cell did not solve date shift.

Table 8.4: Five-day fair image-model comparison. R^2 is the mean of the five date-specific values.

Image model	Mean MAE	Mean RMSE	Mean R^2	Worst RMSE
ResNet50-GRU T7, lens 6	38.72	47.29	-1.385	80.70
ResNet50-GRU T7, mean 1/2/6	38.58	47.43	-1.319	78.30
ResNet50-GRU T7, concat 1/2/6	39.05	47.54	-1.323	82.65
ConvNeXt-tiny-GRU T7, concat	39.63	47.74	-1.329	82.16
ResNet50-LSTM T7, concat	39.95	48.46	-1.394	85.26
Robust ResNet50-GRU T7, concat	40.43	48.53	-1.471	87.19
MobileNetV2-GRU T7, concat	43.04	50.89	-1.742	92.77

Lens 6 was retained for the structured temporal branch because it was a competitive single view, reduced dimensionality and showed the most consistent road-facing content. The attempted five-day random ResNet50-LSTM run contained sequence and audit files but no final trained metrics, so it was marked incomplete rather than inferred.

8.8 Five-day random-sequence diagnostics

Table 8.5: Selected five-day random-sequence results. All have temporal overlap and are not future-date scores.

Base and correction	MAE	RMSE	R^2
Lens-6 ResNet50-GRU T7 + sensor/visual RF	10.105	19.608	0.852
Concat ResNet50-GRU T7 + sensor/visual RF	9.328	19.308	0.856
Concat ResNet50-GRU T9 + sensor/visual RF	8.292	25.251	0.776
ConvNeXt-tiny-GRU T7 + sensor/visual RF	8.951	18.537	0.867
ConvNeXt-tiny-GRU T7 + sensor/AlphaEarth ET	8.825	18.592	0.867
CAMS + ResNet50-GRU T7 + sensor/visual/AE ET	9.856	20.032	0.845
CAMS + same + OSM ET	9.787	19.897	0.847

OSM improved the matched CAMS random result by only 0.134 RMSE and 0.002 R^2 . Five extreme test samples in the final OSM random run had RMSE 188.90 compared with 17.58 for inliers, showing that a small number of peaks dominated the overall squared error.

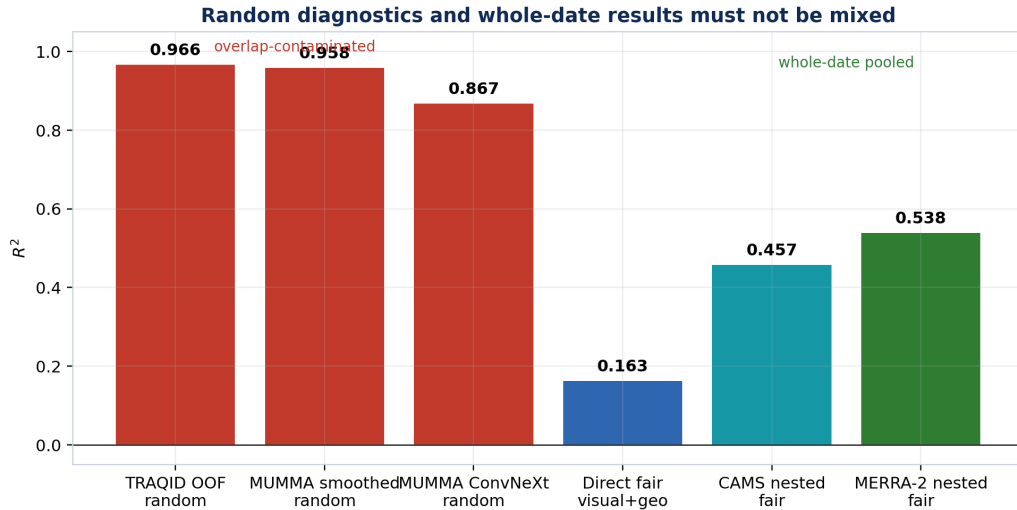


Figure 8.1: Representative random and whole-date pooled results. The bars are displayed together only to show the validation gap; they are not directly comparable claims.

8.9 Five-day direct tabular models

The direct tabular benchmark compared Ridge, Random Forest, ExtraTrees and HistGradient-Boosting across feature groups.

- The best random-row diagnostic used 24 sensor/met/gas/time/mobility features with ExtraTrees: MAE 8.43, RMSE 29.83, R^2 0.730 and Spearman 0.960.
- The best complete fair visual/geospatial row used 192 YOLO/road/OSM/AlphaEarth features with HistGradientBoosting: MAE 34.96, RMSE 47.74, pooled R^2 0.163 and Spearman 0.509.
- Without OSM, visual/road/AlphaEarth Ridge reached RMSE 48.40 and pooled R^2 0.140.

- Nested top-30 feature selection was worse than keeping all 192 features: RMSE 48.87 versus 47.59 and R^2 0.123 versus 0.168.

This result suggests distributed information across correlated feature groups; a hard importance cutoff discarded useful complementary variables.

8.10 Direct image plus residual correction

The best fair direct image-plus-residual configuration was ResNet50-GRU T7 with sensor/AlphaEarth Random Forest correction: MAE 32.82, RMSE 47.33 and pooled R^2 0.187. Sensor/visual correction reached R^2 0.175. ConvNeXt correction reached 0.136, while the robust ResNet variant reached only 0.032. Correction candidates had been compared after observing outer-test results, so this stage was exploratory rather than confirmatory.

8.11 CAMS background plus local increment

The CAMS decomposition was the first major improvement under whole-date evaluation:

Table 8.6: CAMS whole-date local-increment results.

Model	MAE	RMSE	Pooled R^2
Tabular visual/road/AlphaEarth ExtraTrees	27.59	39.59	0.425
Tabular + OSM ExtraTrees	27.52	39.56	0.425
Temporal ResNet50-GRU + sensor/visual RF	26.26	39.89	0.423
Nested ensemble, validation-selected, no OSM	25.281	38.666	0.457
Nested ensemble, fixed 50:50 diagnostic	25.275	38.288	0.468
Nested ensemble, validation-selected, OSM	25.456	38.780	0.454

OSM was useful and interpretable in the standalone tabular branch but slightly worsened the full selected ensemble. The no-OSM validation-selected CAMS model was therefore the earlier frozen candidate.

8.12 MERRA-2 refinement

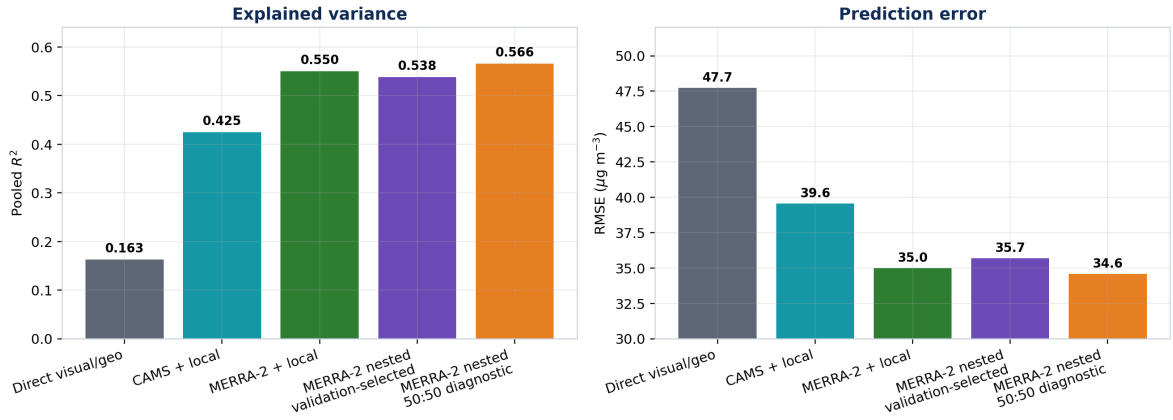
MERRA-2 provided the strongest background signal tested on five-day MUMMA. The standalone MERRA-2 plus YOLO/road/OSM/AlphaEarth ExtraTrees model achieved MAE 23.49, RMSE 35.01 and pooled R^2 0.550 over all 6,873 rows.

The nested ensemble used 6,543 sequence-aligned test rows:

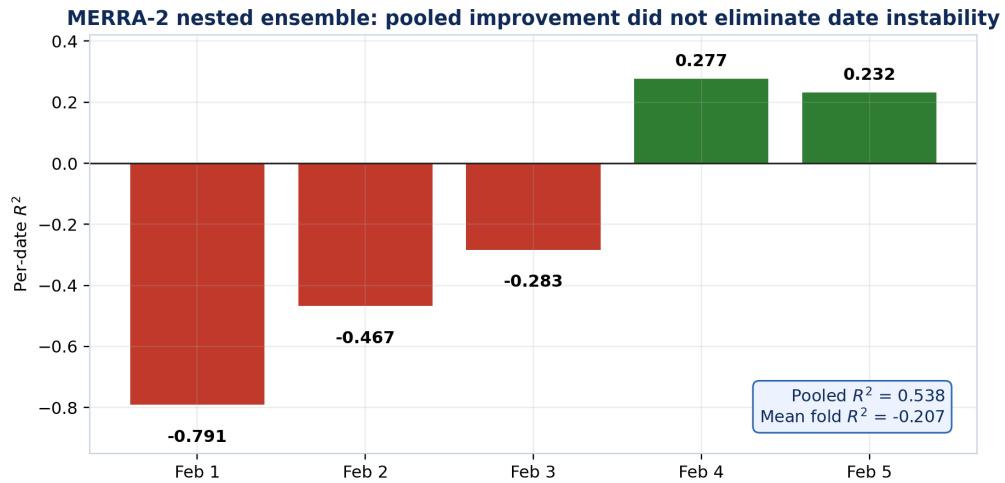
Table 8.7: MERRA-2 nested whole-date ensemble results.

Method	n	MAE	RMSE	Pooled R^2
Fixed equal weight diagnostic	6,543	21.985	34.589	0.566
Tabular branch	6,543	23.520	35.276	0.548
Validation-selected weight	6,543	22.580	35.689	0.538
Temporal branch	6,543	23.650	37.030	0.502

Result. The primary selection-valid MERRA-2 result is MAE 22.58 $\mu\text{g m}^{-3}$, RMSE 35.69 $\mu\text{g m}^{-3}$ and pooled R^2 0.538. The fixed 50:50 result is numerically better but is labelled a diagnostic because that weight was not chosen by the predefined validation rule.

Five-day whole-date results: atmospheric background materially improved pooled performance**Figure 8.2:** Progression from direct whole-date modelling to CAMS and MERRA-2 background-plus-local-increment models.

The selected temporal weights averaged 0.826 and ranged from 0.733 to 0.955. However, the selected model's date-specific R^2 values were -0.791, -0.467, -0.283, 0.277 and 0.232. Its mean fold R^2 was -0.207.

**Figure 8.3:** Date-specific R^2 for the validation-selected MERRA-2 nested ensemble. Positive pooled performance coexists with unstable within-date prediction.

8.13 Atmospheric source ablation

A matched comparison showed:

Table 8.8: Five-day pooled comparison of atmospheric backgrounds with the same local modelling concept.

Background framework	MAE	RMSE	Pooled R^2
Raw MERRA-2 + local	23.484	35.031	0.550
Mean CAMS/MERRA-2 + local	24.872	36.558	0.509
Raw CAMS + local	27.606	39.586	0.425
Nested calibrated atmosphere + local	29.790	41.844	0.357
Nested calibrated atmosphere only	34.305	46.413	0.209

Target-fitted atmospheric calibration did not help and was scientifically less attractive because it introduced dependence on the roadside target. The raw, externally available MERRA-2 background was therefore preferred.

8.14 Architecture 2 results

Architecture 2 combined image and atmospheric features early in one ExtraTrees model. On MUMMA, atmosphere-only and early fusion were almost identical:

Table 8.9: Architecture-2 whole-date results.

Dataset	Variant	MAE	RMSE	Pooled R^2
MUMMA	Atmosphere only	32.288	45.822	0.229
	Early fusion	32.498	45.805	0.230
	Image only	43.289	55.745	-0.141
TRAQID	Atmosphere only	31.200	45.935	0.035
	Early fusion	32.192	46.832	-0.003
	Image only	33.025	49.382	-0.115

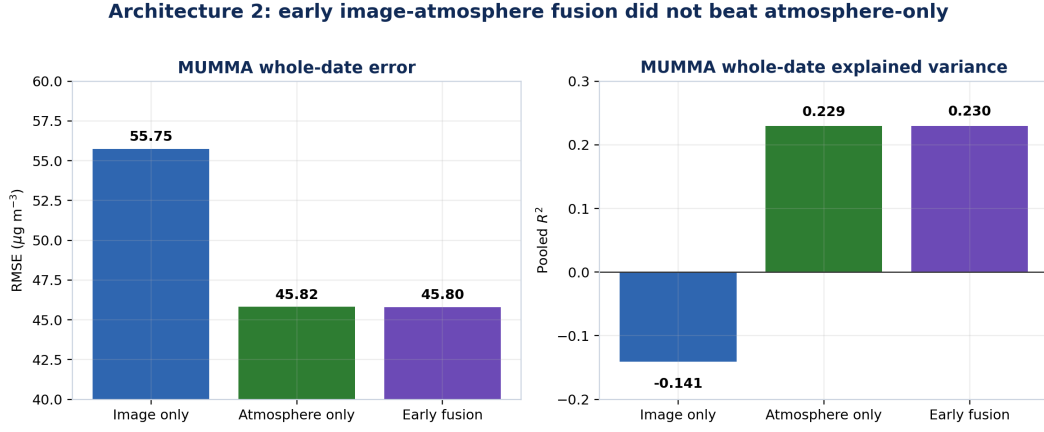


Figure 8.4: MUMMA architecture-2 ablation. The image embedding added almost no value over atmospheric fields in early fusion.

This result does not invalidate images. It shows that a single target-frame PCA embedding concatenated with coarse atmosphere was not the right inductive bias for these five dates. The structured background/local model performed better because it assigned different roles to regional background and local evidence.

8.15 Uncertainty quantification

Validation absolute-residual conformal intervals were added without using outer-test targets for calibration. For the MERRA-2 experiment, the nominal coverage was 90%. Empirical coverage ranged from 0.761 for the temporal branch to 0.850 for the tabular branch; the validation-selected ensemble covered 0.776 with mean interval width $63.14 \mu\text{g m}^{-3}$.

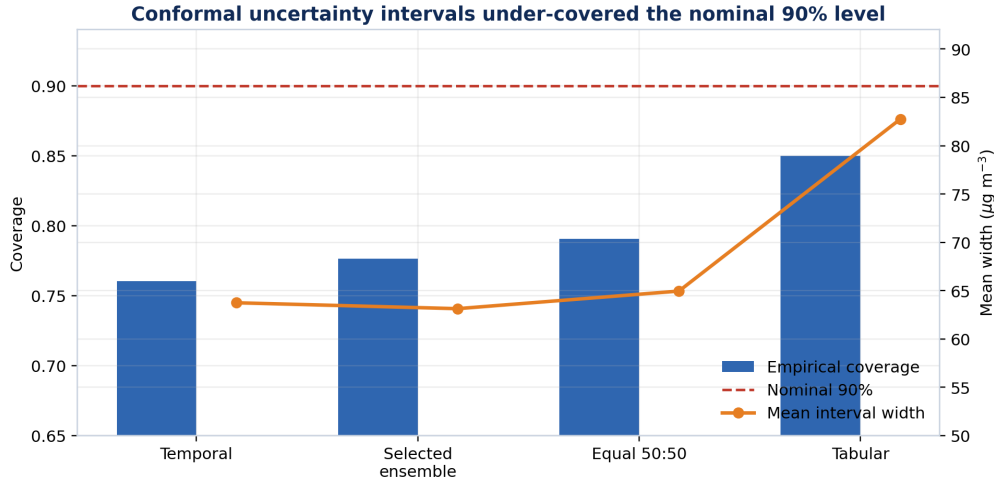


Figure 8.5: MERRA-2 conformal interval coverage and width. Undercoverage indicates residual distribution shift across dates.

The intervals were therefore informative diagnostics but not yet calibrated deployment guarantees. Whole-date distribution shift violated the approximate exchangeability on which simple split-conformal coverage relies.

Feature Relevance and Diagnostics

9.1 Correlation and confounding

Feature relevance was assessed using three complementary views:

1. global Pearson and Spearman correlations;
2. correlations after centring within each date; and
3. held-out-date permutation importance.

The distinction between global and within-date association was essential. Total vehicle count, for example, had global Spearman correlation of approximately -0.179 but within-date correlation of approximately +0.180. This sign reversal is consistent with route and date confounding. Car-heavy segments may have occurred on dates or routes with lower regional background; it does not imply that cars reduce particulate matter.

Repeated preliminary signals included:

- **AlphaEarth:** A52, A51, A06, A49, A10, A14, A36, A39 and A46;
- **road:** lens-6 Laplacian texture, edge density, glare, maximum road area, gray/dry ratio and saturation;
- **traffic:** lens-2 vehicle box area, lens-6 total vehicles, auto-rickshaws, buses, heavy vehicles and exhaust/resuspension proxies;
- **OSM:** commercial count, service roads, bus stops, total road length and hospital count.

Only AlphaEarth A36 was selected in every nested feature-selection fold. Hard top-30 selection still reduced performance, so the final model retained the broader feature representation.

9.2 Spatial variogram

An empirical variogram was computed from all 6,873 PM_{2.5} observations using Haversine distance between latitude/longitude pairs. Within-date semivariance in the first 0–0.5 km bin was 924.13. The analysis showed spatial structure, but repeated routes and temporal evolution were confounded with distance. Consequently, the variogram was treated as a descriptive diagnostic rather than a fitted stationary geostatistical model.

9.3 Sudden-event audit

Thirty-five transitions exceeded the 99.5th percentile absolute-change threshold of $119.74 \text{ } \mu\text{g m}^{-3}$. Twelve were immediate reversals. The maximum recorded `sPM2` was $1,108.64 \text{ } \mu\text{g m}^{-3}$. These events dominate RMSE and may reflect real plumes, sensor lag, clock misalignment, aggregation effects or spikes. The project generated exact raw one-second request windows and video timestamps for all five days so that these cases can be checked against the original stream.

The one-second data should be used for:

- sensor-video lag estimation;
- alternative aggregation windows;
- spike persistence and reversal checks;
- response-time modelling; and
- synchronisation quality control.

It should not be used merely to multiply the number of randomly shuffled, strongly autocorrelated training rows.

9.4 Transfer and calibration experiments

A TRAQID-style weather/time ExtraTrees model transferred poorly to MUMMA: zero-shot MAE 40.62, RMSE 60.23, R^2 -0.332 and Spearman 0.195 over all 6,873 rows. Retraining the same family improved random interpolation but not unseen-date performance. Nested five-date raw weather/time ExtraTrees obtained RMSE 61.44 and R^2 -0.386; affine calibration worsened it. This demonstrated that simple linear recalibration cannot repair a change in the conditional relationship between context and roadside concentration.

Particle Regimes and Source Proxies

10.1 Why the terminology changed

The available sensor table contained cumulative PM mass channels and OPC-like number channels, not chemically speciated reference contributions. Chemical source apportionment normally requires source-discriminating tracers, validated source profiles or differential particle bins. The project therefore adopted three increasingly cautious labels:

1. *latent source-proxy components* for NMF factors;
2. *assumption-conditioned source scenarios* for profile-based NNLS; and
3. *directly derived particle-size regimes* for algebraic mass/number targets.

10.2 Train-partition NMF components

Non-negative matrix factorisation was applied to: `sPM1`, `sPM2`, `sPM4`, `sPM10`, `sNPMp5`, `sNPM1`, `sNPM2`, `sNPM4`, `sNPM10` and `sTPS`. NMF was fitted on the training partition only. Four latent components reconstructed the normalised particle table with RMSE 0.000649 and exact numerical mass closure.

An ExtraTrees context model predicted component fractions from 127 sensor-plus-visual non-PM variables. On the random test split, component R^2 values were 0.508, -3.757, -2.382 and 0.604, with macro fraction MAE 0.0897. The factors are mathematical co-variation patterns. The labels “component 1” through “component 4” deliberately avoid unsupported names such as exhaust or road dust.

10.3 Assumed-profile scenarios

Four illustrative profiles represented vehicle exhaust, road dust, secondary background and unresolved material. Monte Carlo NNLS with 200 draws propagated profile uncertainty. The assumed profile matrix had condition number 126.9 and maximum pairwise cosine similarity 0.970. The median solution assigned 100% to the vehicle scenario, with nearly all other fractions zero.

This apparently decisive solution was actually an identifiability failure. The profiles were too similar relative to the available measurements; the solver selected a boundary solution. The experiment was therefore retained as an assumption-sensitivity demonstration and explicitly rejected as measured source apportionment.

10.4 Derived particle-size regimes

Mass increments and number proxies were used to derive four PM₁₀ mass fractions, an effective diameter, a provisional effective-density proxy and a number proxy. In a random row split, the four mass fractions achieved R^2 near 0.744, effective diameter 0.648, density proxy 0.544 and total-number proxy 0.844. Held-out-date performance collapsed. On the February 1 stress test, representative R^2 values were approximately -72 for a mass fraction, -6.16 for effective diameter, -2.13 for density and -5.12 for number.

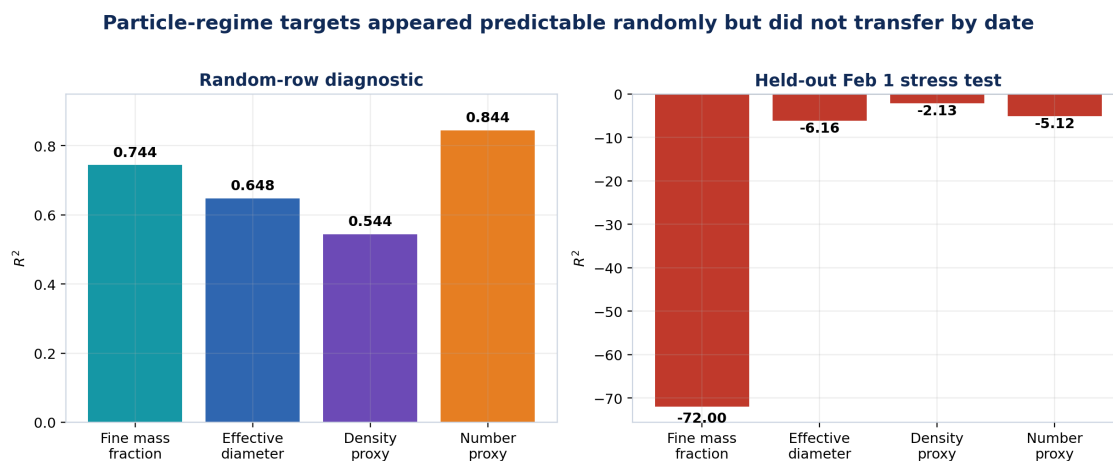


Figure 10.1: Random and held-out-date particle-regime results. The large gap mirrors the validation problem found in PM_{2.5} prediction.

10.5 Requirements for credible source attribution

Future chemical attribution requires:

- raw differential OPC bins with confirmed units and cut diameters;
- independent chemical/speciation measurements such as FTIR or elemental tracers;
- source reference samples or defensible literature profiles;
- uncertainty-aware receptor modelling and identifiability diagnostics; and
- held-out event/date validation.

Until these data exist, traffic, road, OSM and image variables can support predictive association and local-increment modelling, not chemical source percentages.

Two-Month Internship Chronology

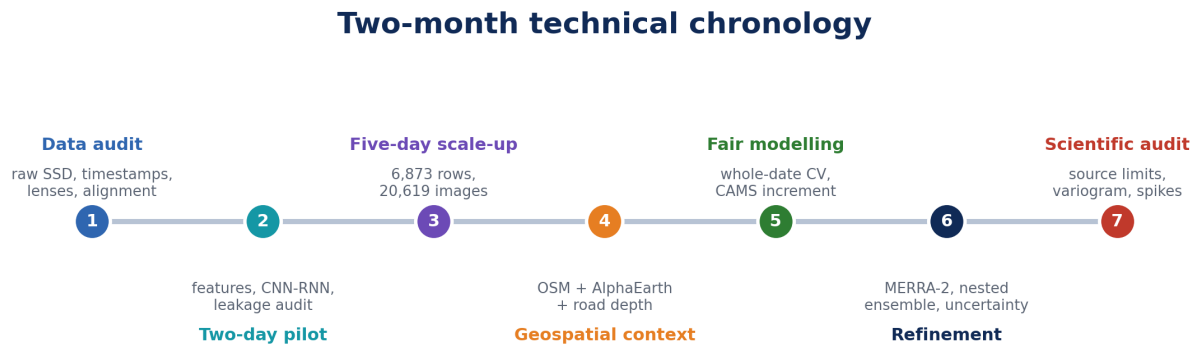


Figure 11.1: Technical progression of the internship. The stages were iterative; later validation findings caused earlier claims to be revised.

11.1 Phase 1: repository and data audit

The first phase identified existing TRAQID code, raw MUMMA folder structure, sensor columns, camera lenses, timestamp conventions and storage constraints. The ingestion tools were designed to read the mounted SSD without modifying it. Configuration files, manifests and artifact directories were used instead of manual file copying.

11.2 Phase 2: two-day pilot

The two-day pilot validated frame extraction, preprocessing, YOLO traffic, SegFormer road features and ResNet50 sequence training. Initial random $T=3$, $T=7$ and $T=9$ experiments revealed strong scores. The overlap audit then showed that these scores were heavily contaminated, motivating a separate fair protocol.

11.3 Phase 3: five-day scale-up

Data for February 2, 4 and 5 were ingested, extracted and preprocessed, then combined with February 1 and 3. The identity audit confirmed 6,873 sensor samples and 20,619 successful frames. YOLO and road features were recomputed for the complete dataset.

11.4 Phase 4: geospatial and atmospheric context

OSM extraction was initially slow because public Overpass queries timed out. A tiled persistent cache and local radius filtering reduced repeated network work; the final run reused 41 of 42 tiles. AlphaEarth embeddings were obtained for all samples. CAMS, ERA5 and MERRA-2 atmospheric products were aligned by time and representative location.

11.5 Phase 5: architecture ladder and fair validation

ResNet50, ConvNeXt-tiny, MobileNetV2, GRU, LSTM, multiple views, robust targets, tabular models and residual correction were compared. Whole-date evaluation showed that none of the direct image variants solved daily distribution shift. The project moved from direct PM_{2.5} prediction to background-plus-local-increment modelling and then to a nested temporal-tabular ensemble.

11.6 Phase 6: refinement, uncertainty and alternative architecture

MERRA-2 replaced CAMS as the strongest tested background. Compact feature budgets, multi-source background combinations, validation-selected and fixed ensemble weights, conformal intervals and target-fitted atmospheric calibration were evaluated. The professor-proposed early image-atmosphere fusion architecture was implemented on both MUMMA and TRAQID and found not to outperform atmosphere-only.

11.7 Phase 7: scientific audit and communication

The final phase consolidated model registries, architecture diagrams, full presentation documentation and careful claims. Feature correlations were reinterpreted in light of date confounding. Variogram and sudden-event analyses were added. Source-apportionment language was revised to reflect the limitations of the measurements.

Discussion

12.1 What worked

The strongest methodological improvement was not a larger CNN. It was the explicit separation of regional background from local roadside increment. Direct image models had to infer both broad daily pollution level and local variation from a short visual sequence. External atmosphere supplied the first part, allowing visual and geospatial variables to focus on the second.

Temporal and tabular branches proved complementary. The GRU captured persistence and scene evolution. The tabular branch exposed traffic counts, road state and slow land-use context. Averaging their errors reduced pooled RMSE. MERRA-2 was more helpful than CAMS in these five dates, but this is an empirical dataset result rather than a universal ranking of the products.

12.2 What did not work

Important negative results were:

- longer overlapping windows inflated some random scores but did not improve whole-date generalisation;
- multi-lens concatenation did not consistently beat lens 6 fairly;
- ConvNeXt, LSTM, MobileNet and robust loss/date balancing did not resolve date shift;
- road depth/metric area was noisy and generally harmful;
- OSM provided small standalone gains but did not improve the selected nested CAMS ensemble;
- top-30 feature selection was worse than all 192 visual/geospatial features;
- affine or target-fitted atmospheric calibration did not repair the generalisation problem;
- early image-atmosphere fusion added almost nothing over atmosphere-only; and
- source-proxy patterns did not generalise by date.

Retaining these results prevents repeated work and supports a more honest final claim.

12.3 Why the final features are not redundant

The professor’s question about redundancy is scientifically important. If road dust and vehicle emissions are already in the atmosphere, why include road and vehicle features? The answer is one of measurement scale and observability:

1. the external background represents a coarse regional state, not the exact roadside plume;
2. raw images are high-dimensional and ambiguous; an ImageNet embedding is not guaranteed to preserve exact vehicle counts or road-surface cues;
3. YOLO and road features provide stable, explicit summaries suited to a small-data tree model;
4. OSM and AlphaEarth describe persistent context not fully visible in one frame; and
5. the model does not add “vehicle PM” and “road PM” as measured masses. It uses these variables jointly to predict one local residual.

Thus, the features are different noisy views of the conditions governing the local increment. Their usefulness must still be established by ablation and held-out data, not by physical storytelling alone.

12.4 Interpretation of current performance

The validation-selected MERRA-2 ensemble explains 53.8% of pooled variance and reduces RMSE to $35.69 \text{ } \mu\text{g m}^{-3}$. This is a meaningful improvement over direct fair models, but it is not a 70–80% generalisation result. The negative mean date-specific R^2 shows that the model is better at capturing differences between days than fine fluctuations within every day.

The fixed 50:50 diagnostic has higher pooled R^2 (0.566), but adopting it after inspecting outer-test performance would constitute test-set model selection. The scientifically defensible primary result remains the validation-selected row. A future-date experiment can fairly compare pre-specified selected and 50:50 policies if both are frozen in advance.

12.5 Threats to validity

- Only five MUMMA dates were available, with unequal routes and sample counts.
- The same dates informed repeated exploratory decisions.
- External backgrounds are model/reanalysis fields, not independent roadside measurements.
- Extreme sensor values may include real events and timing/quality artefacts.
- Image and sensor clocks may have residual lag.
- AlphaEarth dimensions are latent and OSM completeness is location dependent.
- Pooled metrics can conceal poor within-date performance.
- Simple conformal intervals under-covered under date shift.

Conclusions and Future Work

13.1 Conclusions

This internship transformed a visually impressive but overlap-sensitive image-sequence benchmark into a multimodal, leakage-audited research pipeline. The work showed that:

1. random-window scores as high as 0.95–0.97 can be reproduced but are not evidence of unseen-date generalisation;
2. direct CNN-RNN and tabular models struggle under whole-date shift;
3. separating external atmospheric background from local roadside increment materially improves fair pooled performance;
4. MERRA-2 was the strongest tested background on five-day MUMMA;
5. temporal and tabular local branches provide complementary predictions;
6. current pooled performance remains accompanied by negative mean per-date R^2 ; and
7. chemical source attribution is not supported by the current sensor channels.

13.2 Frozen candidate for future-date confirmation

The recommended confirmatory pipeline is:

1. align and preprocess lens-6 frames at the sensor timestamps;
2. encode frames with frozen ResNet50 and build T=7 sequences within `run_id`;
3. retrieve MERRA-2 background without fitting it to roadside targets;
4. train the GRU on the local increment;
5. cross-fit a sensor/YOLO/road Random Forest residual correction by run;
6. train a YOLO/road/OSM/AlphaEarth ExtraTrees local branch;
7. choose the convex temporal/tabular weight on validation predictions only;
8. calibrate an uncertainty interval on validation residuals;
9. freeze preprocessing, features, architecture, hyperparameters and weights; and
10. evaluate once on untouched future dates.

13.3 Priority future experiments

1. Collect new dates spanning different routes, weather, traffic and pollution regimes.
2. Obtain raw one-second streams and estimate sensor-video lag before retraining.
3. Add independent regulatory or local background monitors where possible to evaluate and bias-correct atmospheric products without using the roadside test target.
4. Use group-aware domain adaptation only within nested training data.
5. Model extreme events separately or use mixture/quantile objectives.
6. Recalibrate uncertainty using leave-one-date-out or weighted conformal methods designed for distribution shift.
7. Confirm OPC units and obtain differential bins plus chemical reference measurements before restarting source attribution.

13.4 Final scientific statement

The project has identified a promising and reproducible approach, not a deployment-ready universal predictor. Its most important outcome is the combination of engineering completeness and scientific restraint: a full multimodal pipeline, an improved background-aware model, explicit leakage audits, transparent negative results and a clear decisive next experiment.

Reproducibility and Artifact Index

The complete source code, experiment configurations, tests, documentation and report sources are available in the public GitHub repository [Roadside PM Visual Intelligence](#). Large raw datasets, extracted frames, cached geospatial responses, learned embeddings and trained model binaries are intentionally excluded from Git because of size, licensing and data-governance constraints. The local artifact paths below document exact provenance and can be regenerated using the version-controlled pipelines.

A.1 Canonical data products

Artifact	Repository path
Five-day sensor table	<code>artifacts/runs/mumma_5day_combined_lenses_1_2_6_v1/sensor_10s.csv</code>
Five-day frame manifest	<code>artifacts/runs/mumma_5day_combined_lenses_1_2_6_v1/manifests/preprocessed_frames.csv</code>
YOLO frame features	<code>artifacts/runs/mumma_5day_visual_features_v1/yolo_frame_features.csv</code>
YOLO objects	<code>artifacts/runs/mumma_5day_visual_features_v1/yolo_objects.csv</code>
Road features	<code>artifacts/runs/mumma_5day_visual_features_v1/road_features_lenses_1_6.csv</code>
OSM 250 m context	<code>artifacts/runs/mumma_5day_geospatial_corridor_v1/osm_250m.csv</code>
AlphaEarth embeddings	<code>artifacts/runs/mumma_5day_geospatial_alphaearth_v1/alphaearth_2025_point.csv</code>
Atmospheric background	<code>artifacts/runs/mumma_5day_atmospheric_background_v1/background_hourly.csv</code>
Canonical non-PM modelling table	<code>artifacts/runs/mumma_5day_complete_nonpm_models_v1/modeling_table.csv</code>
Feature groups	<code>artifacts/runs/mumma_5day_complete_nonpm_models_v1/feature_groups.json</code>

A.2 Principal model artifacts

Experiment	Repository path
Five-day fair image ladder	artifacts/runs/mumma_5day_image_ladder_T7_gru_v1
Direct complete tabular benchmark	artifacts/runs/mumma_5day_complete_nonpm_models_v1
CAMS local increment	artifacts/runs/mumma_5day_background_local_increment_osm_alphaearth_v1
CAMS nested ensemble	artifacts/runs/mumma_5day_nested_background_ensemble_fair_v1
CAMS nested ensemble with OSM	artifacts/runs/mumma_5day_nested_background_ensemble_osm_fair_v1
MERRA-2 local increment	artifacts/runs/mumma_5day_merra2_local_osm_alphaearth_fair_v1
MERRA-2 nested ensemble	artifacts/runs/mumma_5day_nested_merra2_ensemble_osm_fair_v1
MERRA-2 ensemble + uncertainty	artifacts/runs/mumma_5day_nested_merra2_ensemble_osm_uncertainty_fair_v1
Architecture-2 MUMMA	artifacts/runs/mumma_5day_architecture2_stage1_fusion_fair_v1
Architecture-2 TRAQID	artifacts/runs/traqid_architecture2_stage1_fusion_fair_v1
TRAQID matched random backgrounds	artifacts/runs/traqid_T7_background_random_comparison_v1
TRAQID nested backgrounds	artifacts/runs/traqid_T7_nested_background_ensemble_fair_v1

A.3 Project documentation

- docs/pm25_experiment_consolidation.md: audited PM modelling story.
- docs/model_architecture_catalog.md: architecture-by-architecture code and status catalogue.
- docs/pm25_model_registry.csv: exact PM result index.
- docs/model_architecture_registry.csv: architecture registry.
- docs/experiment_registry.csv: run-level experiment inventory.
- docs/complete_project_export_for_presentation.md: broad project export used for presentation development.
- docs/data_lineage.md: dataset lineage and processing paths.

Detailed Feature Dictionary

B.1 YOLO feature semantics

Category	Representative variables	Interpretation
Vehicle totals	total, motor and heavy vehicle counts	Instantaneous traffic activity and fleet composition.
Classes	car, motorcycle, auto-rickshaw, bus, truck, bicycle, fallback	Explicit mix of visually detected road users.
Confidence	average and maximum confidence	Detection certainty and scene difficulty; not an emission variable.
Box geometry	mean/max box area and occupancy	Proxy for proximity and visual scene occupancy.
Exhaust proxy	class-weighted initial exhaust score	Heuristic source opportunity; not measured exhaust mass.
Resuspension proxy	class/size-weighted road activity score	Heuristic opportunity for mechanically generated dust.
Per-lens variants	lens 1, 2 and 6 copies	Preserves view-specific traffic information rather than averaging all cameras blindly.
Availability	yolo_lens_count	Quality-control indicator for available views.

B.2 Road feature semantics

Variable	Interpretation
Road area ratio	Fraction of the image classified as road.
Mean brightness	Illumination of the road mask.
Mean saturation	Colour intensity, potentially affected by surface and lighting.
Contrast standard deviation	Variation in road appearance.

Variable	Interpretation
Shadow/glare ratios	Illumination artefacts and surface reflectance.
Brown-pixel ratio	Heuristic colour proxy for dusty/bare-looking road.
Gray/dry-pixel ratio	Heuristic dry pavement appearance.
Edge density	Fine spatial structure and markings/texture.
Laplacian standard deviation	High-frequency texture/sharpness.
Haze-flatness proxy	Low-contrast appearance proxy; not direct aerosol optical depth.

B.3 OSM feature semantics

Feature group	Context represented
Fuel stations, bus stops, parking	Traffic generation, stopping and idling opportunities.
Restaurants, marketplaces, commercial, retail	Human activity, delivery, parking and congestion context.
Construction, industrial, factory, warehouse	Heavy-vehicle, loading and dust-source opportunity.
Schools, hospitals, parks	Land-use and activity context.
Road count and total length	Local network density.
Motorway through service/living street counts	Road hierarchy and likely traffic function.

Model and Experiment Status

Experiment family	Status	Main conclusion
TRAQID OOF clipped residual	Complete, random diagnostic	Highest historical R^2 0.966; overlap-contaminated.
MUMMA two-day sequence ladder	Complete, exploratory	Longer T improved random score and leakage simultaneously.
Five-day frozen-CNN ladder	Complete, fair	Backbone changes did not solve date shift.
Five-day LSTM random run	Incomplete	Sequences/audit exist; no final training metrics found.
Five-day direct tabular	Complete	Random interpolation strong; fair visual/geospatial R^2 about 0.16.
Depth road area	Complete, rejected feature	Provisional values were heavily flagged and generally harmful.
OSM	Complete	Useful context; small matched gain; no full-ensemble gain.
AlphaEarth	Complete	Distributed latent signal; top-30 selection harmful.
CAMS local increment	Complete	Major fair improvement to pooled R^2 about 0.425–0.457.
MERRA-2 local increment	Complete	Strongest tested five-day background; pooled R^2 0.550 standalone.
MERRA-2 nested ensemble	Complete, exploratory candidate	Selected pooled R^2 0.538; equal-weight diagnostic 0.566.
Conformal uncertainty	Complete, diagnostic	Intervals under-covered under date shift.
Architecture 2 early fusion	Complete on both datasets	Did not beat atmosphere-only.
NMF source proxies	Complete, exploratory	Some random component predictability; no chemical identities.
Assumed source scenarios	Complete, rejected inference	Degenerate, ill-conditioned solution.
Particle regimes	Complete, exploratory	Strong random results; poor held-out-date transfer.

Experiment family			Status	Main conclusion		
Variogram events	and	sudden	Complete, diagnos- tic	Spatial/time extreme-event identified.	confounding validation	and needs

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